

‘what customers want’ can guide us to make the correct choices for the product.

Connectivity is more challenging

Connectivity is far more complex than simply adding a motor or light to a product.

You don’t control the ecosystem.

Connectivity means your device is communicating with something else, usually something outside your control. That might be a phone, a router, or another third-party device. You can test and optimise your system, but you can’t guarantee what the other end looks like. That’s a hard design constraint.

You don’t control the environment. Where will your product be used? In a concrete apartment? In a crowded café filled with RF noise? In a quiet suburban garage? You don’t know—and you can’t assume a clean environment. Signal attenuation, interference, and real-world variability are inevitable.

Design for reality, not the lab. Products face real-world unpredictability, so connectivity solutions must prioritise robustness, error handling, fallback mechanisms, and user experience, alongside RF range and bandwidth.

In connected vehicles, real-world experiences shape the understanding of the system. For example, a Reddit user shared an issue where TPMS sensors froze, causing their phone key to lose Bluetooth functionality but still handle data. This scenario provides valuable insights for engineers to enhance system design and prevent similar issues.

Modern vehicles rely heavily on wireless connectivity, encompassing Bluetooth, Wi-Fi, cellular, UWB, and BLE (Bluetooth Low Energy), to enable a range of features. In this case, we are looking at:

- TPMS (tire pressure monitoring

The hidden pitfalls of connected devices—and how smart design can avoid them

These takeaways underscore that in the rapidly evolving world of connected electronics, robust security, real-world performance, and user-centric design are non-negotiable for successful products.

- ***Technology on paper vs product reality.*** A technology that appears robust in theory may encounter significant hurdles in real-world product integration, particularly in terms of user experience, compatibility, and cost.
- ***Customer-centric product development.*** Successful connectivity products must prioritise reliability, seamless operation, security, and affordability—mirroring the dependability of mechanical keys without added complexity or cost.
- ***Smartphone integration challenges.*** Transitioning vehicle keys to smartphones introduces new obstacles, such as ensuring consistent performance across platforms and keeping up with updates and changes on multiple platforms.
- ***Environmental and ecosystem factors are critical.*** Connectivity solutions must account for unpredictable real-world environments, as signal range can drop dramatically due to obstacles like walls, other vehicles, or even the human body, requiring robust design margins.
- ***Security must be built-in, not bolted on.*** High-profile attacks, such as remote vehicle takeovers and medical device recalls, highlight the need for security to be integral to the design process, ensuring proper isolation, authentication, and encryption from the outset.
- ***Ecosystem complexity increases attack surface.*** Modern vehicles and IoT products connect via multiple wireless protocols (Wi-Fi, BLE, cellular, proprietary RF), each introducing potential vulnerabilities. A single weak link can compromise the entire system.
- ***Lessons from medical device recalls.*** Basic design lapses, such as missing authentication and unencrypted data, can have life-threatening consequences, reinforcing the need for rigorous security hygiene in all connected products.
- ***Proactive design prevents costly failures.*** Addressing connectivity and security challenges early in the design phase, rather than reacting to field failures, saves time, money, and reputation.

system)—typically connects to a central BLE hub.

- Phone as a key (PaaK)—typically BLE-based.
- Bluetooth audio—uses classic Bluetooth (not BLE).

The user noted that Bluetooth audio worked, but BLE-based functions, such as the TPMS and phone-as-a-key, did not. This narrows the issue to the vehicle’s BLE stack or communication path, which is usually separate from the classic Bluetooth path. Several factors that may have contributed to these symptoms are discussed further.

BLE operates in the 2.4GHz ISM band—a frequency highly susceptible to interference and signal attenuation. In ideal conditions, a smartphone and vehicle BLE system

can connect over distances exceeding 50 metres. However, in real world scenarios, obstacles and body interference can dramatically reduce this range.

In an open-line-of-sight scenario, BLE can achieve a range of over 50 metres, depending on the implementation. However, introducing obstacles like a car and a wall can drastically reduce that range to around 10 metres. Furthermore, if the phone is placed in a back pocket, the human body alone can attenuate the signal strength by 12-15dB, potentially reducing the range to 3 metres or even less. This highlights the practical complexity engineers must account for—while BLE is highly efficient, its effective range is extremely variable and